
UTM
UNIVERSITI TEKNOLOGI MALAYSIA
INSPIRING CREATIVE AND INNOVATIVE MINDS
www.utm.my
System of Linear Equations
& Matrices


UTM
UNIVERSITI TEKNOLOGI MALAYSIA
INSPIRING CREATIVE AND INNOVATIVE MINDS
www.utm.my
PART A

UTM UNIVERSITI TEKNOLOGI MALAYSIA

www.utm.my

Diagnostic Test

$$A=\begin{bmatrix}2 & 1 & 0\\ 5 & -1 & 3\end{bmatrix} \quad B=\begin{bmatrix}4 & 0 & -2\\ 7 & 11 & 8\end{bmatrix} \quad C=\begin{bmatrix}2 & 1\\ 0 & 4\\ 1 & 3\end{bmatrix}$$

(a)The (1,3) entry of B is ?
(b)Size of matrix C
(c) A + B **(d) 3A-B** **(e)AC**
(f)A^T+C **(g)CB**

UTM UNIVERSITI TEKNOLOGI MALAYSIA

www.utm.my

Linear Equations

- An equation of the form

$$a_1x_1 + a_2x_2 + \ldots\ldots + a_nx_n = b$$

is called a linear equation in variables

$$x_1, x_2, \ldots, x_n$$

INSPIRING CREATIVE AND INNOVATIVE MINDS

Linear Equations

$$a_1, a_2, \ldots, a_n$$

- real numbers, called the coefficients of the variables

$$x_1, x_2, \ldots, x_n$$

- The number *b* is called the constant term of the equation.

UTM UNIVERSITI TEKNOLOGI MALAYSIA
www.utm.my

Linear Equations

- The column matrix

$$X = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} = \begin{bmatrix} x_1 & x_2 & .. & x_n \end{bmatrix}^T$$

is called the matrix of variables.

INSPIRING CREATIVE AND INNOVATIVE MINDS

Linear Equations

- A set of numbers

$$s_1, s_2, \ldots, s_n$$

is called a solution to the linear equation if.

$$a_1 s_1 + a_2 s_2 + \ldots\ldots + a_n s_n = b$$

$$X = \begin{bmatrix} s_1 & s_2 & .. & s_n \end{bmatrix}^T$$

(values s1,s2,…..s*n* are substitute for *x*1,*x*2,….*xn*)

UTM UNIVERSITI TEKNOLOGI MALAYSIA
www.utm.my

example

- Show that

$X=[1\ -2]^T$

is a solution to the equation

$2x_1 - 3x_2=8$

but that

$Y=[1\ \ 1]^T$

is not a solution.

INSPIRING CREATIVE AND INNOVATIVE MINDS

UTM UNIVERSITI TEKNOLOGI MALAYSIA

www.utm.my

example

- $X=[1\ -2]^T$ is a solution because $x_1=1$ and $x_2=-2$ satisfy the equation:

$$2(1) - 3(2) = 8$$

but $x_1=1$and $x_2=1$ do not satisfy the equation:

$$2(1) - 3(1) \neq 8$$

so $[1\ \ 1]^T$ is not a solution.

UTM UNIVERSITI TEKNOLOGI MALAYSIA

www.utm.my

Example

- Determine whether either of the points (–1, –5) and (0, –2) is a solution to the given system of equations.

$$y = 3x - 2$$
$$y = -x - 6$$

Answer

To check the given possible solutions, just plug the *x*- and *y*-coordinates into the equations, and check to see if they work.

INSPIRING CREATIVE AND INNOVATIVE MINDS

Example

checking (–1, –5):

$$(-5) = 3(-1) - 2$$
$$-5 = -3 - 2$$
$$-5 = -5 \quad \text{(solution checks)}$$

$$(-5) = -(-1) - 6$$
$$-5 = 1 - 6$$
$$-5 = -5 \quad \text{(solution checks)}$$

Since the given point works in each equation, it is a solution to the system.

UTM UNIVERSITI TEKNOLOGI MALAYSIA
www.utm.my

Example

checking (0, –2):

$$(-2) = 3(0) - 2$$
$$-2 = 0 - 2$$
$$-2 = -2 \quad \text{(solution checks)}$$

$$(-2) = -(0) - 6$$
$$-2 = 0 - 6$$
$$-2 = -6 \quad (-2 \neq -6)$$

Therefore, only the point (–1, –5) is a solution to the system

UTM UNIVERSITI TEKNOLOGI MALAYSIA

www.utm.my

System of Linear Equations

- A finite collection of linear equations is called a system of linear equations.
- A system of *m* equations in *n* variables,

$$
\begin{aligned}
a_{11}x_1 + a_{12}x_2 + ... + a_{1n}x_n &= b_1 \\
a_{21}x_1 + a_{22}x_2 + ... + a_{2n}x_n &= b_2 \\
\vdots \qquad & \vdots \qquad \vdots \\
a_{m1}x_1 + a_{m2}x_2 + ... + a_{mn}x_n &= b_m
\end{aligned}
$$

UTM UNIVERSITI TEKNOLOGI MALAYSIA

www.utm.my

System of Linear Equations

- A system has a solution if there exist numbers s_1, s_2, …s_n which satisfy each of the equations simultaneously.
- The system is called consistent if it has one or more solutions (unique or infinite solutions)
- The system is called inconsistent if it has no solution.

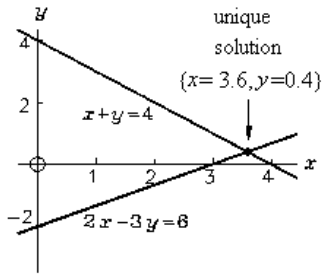
UTM UNIVERSITI TEKNOLOGI MALAYSIA

www.utm.my

Unique Solution

- A system,

$$\begin{cases} 1x + 1y = 4 \\ 2x - 3y = 6 \end{cases}$$



- The point where the lines cross {x=3.6, y=0.4} is the solution that satisfies both equations simultaneously.
- The solution is unique and consistent

UTM UNIVERSITI TEKNOLOGI MALAYSIA

www.utm.my

example

- A system,

$$x_1 + x_2 = 5 \quad (1)$$
$$3x_1 + 2x_2 = 12 \quad (2)$$

- Has a solution, x_1=2 and x_2=3
- 2 lines are not parallel and intersect at point (2,3).
- Consistent

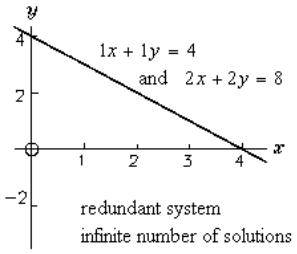
UTM UNIVERSITI TEKNOLOGI MALAYSIA

www.utm.my

Infinite Many Solutions

- A systems,

$$\begin{cases} 1x + 1y = 4 \\ 2x + 2y = 8 \end{cases}$$



- This system is redundant because the second equation is equivalent to the first one.
- They 'cross' at an infinite number of points, so there are an infinite number of solutions
- Consistent

UTM UNIVERSITI TEKNOLOGI MALAYSIA

www.utm.my

example

- A system,

$$x_1 + 2x_3 = 6 \quad (1)$$
$$x_2 + x_3 = 2 \quad (2)$$

- Has many solutions, $x_1=6-2t$, $x_2=2-t$ and $x_3=t$ where t is arbitrary.

UTM UNIVERSITI TEKNOLOGI MALAYSIA

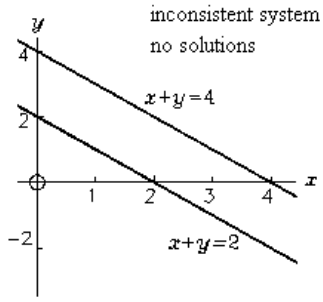
www.utm.my

No Solutions

- A system

$$\begin{cases} x + y = 2 & (1) \\ x + y = 4 & (2) \end{cases}$$

- Consists of two parallel lines that never cross. Thus there is no solution.



- Inconsistent

UTM UNIVERSITI TEKNOLOGI MALAYSIA

www.utm.my

example

- A system,

$$3x_1 + x_2 = 5 \quad (1)$$
$$3x_1 + x_2 = 7 \quad (2)$$

- (2)-(1), 0=2 (no solution)

- 2 lines are parallel and do not intersect.
- Inconsistent

INSPIRING CREATIVE AND INNOVATIVE MINDS



UTM
UNIVERSITI TEKNOLOGI MALAYSIA

Matrices

www.utm.my

- A rectangular array of number is called a **matrix**, and the numbers them selves are called the **entries** of the matrix.
- Matrices are usually denoted by upper case letter *A*, *B*, *C*, etc.

UTM UNIVERSITI TEKNOLOGI MALAYSIA

www.utm.my

example

- The following are matrices:

$$A = \begin{bmatrix} 2 & 0 & -4 \\ 3 & 1 & 7 \end{bmatrix} \qquad B = \begin{bmatrix} 1 \\ 4 \end{bmatrix}$$

$$C = \begin{bmatrix} -1 & 5 & 2 \end{bmatrix} \qquad D = \begin{bmatrix} 4 & 12 & 6 \\ 7 & 0 & 4 \\ 11 & 3 & -5 \end{bmatrix}$$

UTM UNIVERSITI TEKNOLOGI MALAYSIA

www.utm.my

Matrices

- Matrices come in various (rectangular) shapes depending on the number of **rows** and **columns**.
- A matrix with **m rows** and **n columns** is said to have **size m×n**.

INSPIRING CREATIVE AND INNOVATIVE MINDS

UTM UNIVERSITI TEKNOLOGI MALAYSIA

www.utm.my

example

- 2×3 matrix

$$A = \begin{bmatrix} 2 & 0 & -4 \\ 3 & 1 & 7 \end{bmatrix}$$

- 3×3 matrix

$$D = \begin{bmatrix} 4 & 12 & 6 \\ 7 & 0 & 4 \\ 11 & 3 & -5 \end{bmatrix}$$

UTM UNIVERSITI TEKNOLOGI MALAYSIA

www.utm.my

Matrices

- A matrix of size **1×n** is called a **row matrix**.
- A matrix of size **m×1** is called a **column matrix**.
- A matrix with equal numbers of rows and columns is called a **square matrix**.

INSPIRING CREATIVE AND INNOVATIVE MINDS

UTM UNIVERSITI TEKNOLOGI MALAYSIA

www.utm.my

example

- Column matrix, $B = \begin{bmatrix} 1 \\ 4 \end{bmatrix}$
- Row matrix, $C = \begin{bmatrix} -1 & 5 & 2 \end{bmatrix}$
- Square matrix, $D = \begin{bmatrix} 4 & 12 & 6 \\ 7 & 0 & 4 \\ 11 & 3 & -5 \end{bmatrix}$

UTM UNIVERSITI TEKNOLOGI MALAYSIA

www.utm.my

Matrices

- The entry in row i and column j is called the (i,j)-entry of the matrix.
- Example:

  The (1,2) entry of $A = \begin{bmatrix} 2 & 0 & -4 \\ 3 & 1 & 7 \end{bmatrix}$

  is 0, while the (2,3) entry is 7. ($a_{1,2}$= 0 , $a_{2,3}$= 7)

UTM UNIVERSITI TEKNOLOGI MALAYSIA

www.utm.my

Matrices

- If A is an m×n matrix and if the (i,j)-entry of A is denoted a_{ij}, then A is displayed as follows:

$$A = [a_{ij}] = \begin{bmatrix} a_{11} & a_{12} & a_{13} & \dots & a_{1n} \\ a_{21} & a_{22} & a_{23} & \dots & a_{2n} \\ \vdots & \vdots & \vdots & \dots & \vdots \\ a_{m1} & a_{m2} & a_{m3} & \dots & a_{mn} \end{bmatrix}$$

- a_{ij} is the entry in row i and column j .

UTM UNIVERSITI TEKNOLOGI MALAYSIA

www.utm.my

Example

- Example:

$$B = \begin{bmatrix} 5 & 2 & 8 \\ 7 & 4 & 3 \\ 3 & 9 & 6 \end{bmatrix}$$

The (3,1) entry of B is ?

$b_{1,3}$= ?

INSPIRING CREATIVE AND INNOVATIVE MINDS

UTM UNIVERSITI TEKNOLOGI MALAYSIA

www.utm.my

Exercise

$$X = \begin{bmatrix} 0 & 5 & 3 & 15 \\ 9 & 6 & 45 & 34 \\ 23 & -3 & 30 & 50 \\ 89 & 64 & 0 & 90 \end{bmatrix}$$

- What is the (3,4)-entry?

UTM UNIVERSITI TEKNOLOGI MALAYSIA

www.utm.my

Matrices

- The diagonal entries in an m×n matrix A=[a_{ij}] are a_{11}, a_{22} , a_{33},… and they form the **main diagonal** of A.

$$A = [a_{ij}] = \begin{bmatrix} a_{11} & a_{12} & a_{13} & \cdots & a_{1n} \\ a_{21} & a_{22} & a_{23} & \cdots & a_{2n} \\ \vdots & \vdots & \vdots & \cdots & \vdots \\ a_{m1} & a_{m2} & a_{m3} & \cdots & a_{mn} \end{bmatrix}$$

UTM UNIVERSITI TEKNOLOGI MALAYSIA

www.utm.my

Matrices

- A **diagonal matrix** is a square matrix whose nondiagonal entries are zero.
- Example: n×n **identity matrix** I_n.

$$I_4 = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

UTM UNIVERSITI TEKNOLOGI MALAYSIA

www.utm.my

Matrices

- The m×n matrix whose entries are all zero is called the **zero matrix** and will be denoted by 0.

$$\begin{bmatrix} 0 & 0 & 0 & \dots & 0 \\ 0 & 0 & 0 & \dots & 0 \\ \vdots & \vdots & \vdots & \dots & \vdots \\ 0 & 0 & 0 & \dots & 0 \end{bmatrix}$$

UTM UNIVERSITI TEKNOLOGI MALAYSIA

www.utm.my

Matrices

- Two matrices A and B are equal (A=B) if they have the same number of rows and columns and if corresponding entries are equal.
- If $A=[a_{ij}]$ and $B=[b_{ij}]$ are both $m \times n$,

  $[a_{ij}] = [b_{ij}]$ if and only if $a_{ij} = b_{ij}$ for all i and j.

UTM UNIVERSITI TEKNOLOGI MALAYSIA

www.utm.my

example

- Given

$$A = \begin{bmatrix} 2 & 0 & -4 \\ 3 & 1 & 7 \end{bmatrix} \qquad B = \begin{bmatrix} 2 & 0 \\ 3 & 1 \end{bmatrix} \qquad C = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

- Discuss the possibility that *A*=*B*, *A*=*C* and *B*=*C*.

INSPIRING CREATIVE AND INNOVATIVE MINDS

UTM UNIVERSITI TEKNOLOGI MALAYSIA

www.utm.my

example

- Given

$$A=\begin{bmatrix}2 & 0 & -4\\ 3 & 1 & 7\end{bmatrix} \qquad B=\begin{bmatrix}2 & 0\\ 3 & 1\end{bmatrix} \qquad C=\begin{bmatrix}a & b\\ c & d\end{bmatrix}$$

- $A=B$, impossible because A and B are of different sizes ($A\neq B$)
- Similarly $A\neq C$.

UTM UNIVERSITI TEKNOLOGI MALAYSIA

www.utm.my

example

$$B=\begin{bmatrix}2 & 0\\ 3 & 1\end{bmatrix} \qquad C=\begin{bmatrix}a & b\\ c & d\end{bmatrix}$$

- $B=C$, is possible provided that corresponding entries are equal.
- Only if a=2, b=0, c=3, d=1.

UTM UNIVERSITI TEKNOLOGI MALAYSIA

www.utm.my

Matrix Addition

- If A and B are 2 matrices of the same size, A+B is defined to be the matrix of the same size formed by adding corresponding entries.
- If $A=[a_{ij}]$ and $B=[b_{ij}]$,

$$[a_{ij}]+[b_{ij}] =[a_{ij}+b_{ij}]$$

UTM UNIVERSITI TEKNOLOGI MALAYSIA

www.utm.my

Matrix Addition

- A–B is defined by subtracting corresponding entries.
- If $A=[a_{ij}]$ and $B=[b_{ij}]$,

$$[a_{ij}] -[b_{ij}] =[a_{ij}-b_{ij}]$$

INSPIRING CREATIVE AND INNOVATIVE MINDS

UTM UNIVERSITI TEKNOLOGI MALAYSIA

www.utm.my

Matrix Addition

$$A = \begin{pmatrix} a_{1,1} & a_{1,2} & a_{1,3} \\ a_{2,1} & a_{2,2} & a_{2,3} \end{pmatrix} \qquad B = \begin{pmatrix} b_{1,1} & b_{1,2} & b_{1,3} \\ b_{2,1} & b_{2,2} & b_{2,3} \end{pmatrix}$$

$$A+B = \begin{pmatrix} a_{1,1}+b_{1,1} & a_{1,2}+b_{1,2} & a_{1,3}+b_{1,3} \\ a_{2,1}+b_{2,1} & a_{2,2}+b_{2,2} & a_{2,3}+b_{2,3} \end{pmatrix}$$

UTM UNIVERSITI TEKNOLOGI MALAYSIA

www.utm.my

Matrix Addition

$$A = \begin{pmatrix} a_{1,1} & a_{1,2} & a_{1,3} \\ a_{2,1} & a_{2,2} & a_{2,3} \end{pmatrix} \qquad B = \begin{pmatrix} b_{1,1} & b_{1,2} & b_{1,3} \\ b_{2,1} & b_{2,2} & b_{2,3} \end{pmatrix}$$

$$A-B = \begin{pmatrix} a_{1,1}- b_{1,1} & a_{1,2}- b_{1,2} & a_{1,3}- b_{1,3} \\ a_{2,1}- b_{2,1} & a_{2,2}- b_{2,2} & a_{2,3}- b_{2,3} \end{pmatrix}$$

INSPIRING CREATIVE AND INNOVATIVE MINDS

example

$$A=\begin{bmatrix}3 & 2 & 0\\ 5 & 1 & -4\end{bmatrix} \qquad B=\begin{bmatrix}-1 & 4 & 2\\ 7 & -5 & 3\end{bmatrix}$$

$$A+B=\begin{bmatrix}3+(-1) & 2+4 & 0+2\\ 5+7 & 1+(-5) & -4+3\end{bmatrix}=\begin{bmatrix}2 & 6 & 2\\ 12 & -4 & -1\end{bmatrix}$$

$$A-B=\begin{bmatrix}3-(-1) & 2-4 & 0-2\\ 5-7 & 1-(-5) & -4-3\end{bmatrix}=\begin{bmatrix}4 & -2 & -2\\ -2 & 6 & -7\end{bmatrix}$$

UTM UNIVERSITI TEKNOLOGI MALAYSIA
www.utm.my

Theorem

- If A, B and C denote arbitrary m×n matrices, then
  - A+B = B+A (commutative law)
  - A+(B+C)=(A+B)+C (associative law)
  - 0+A=A (0 is the m×n zero matrix)
  - A+(–A)=0

INSPIRING CREATIVE AND INNOVATIVE MINDS

UTM UNIVERSITI TEKNOLOGI MALAYSIA

www.utm.my

Scalar Multiplication

- If A is a matrix and c is a number, the **scalar product cA** is the matrix formed from A by multiplying each entry of A by the number c.
- If $A=[a_{ij}]$

$$cA=c[a_{ij}]$$

UTM UNIVERSITI TEKNOLOGI MALAYSIA

www.utm.my

Scalar Multiplication

$$A=\begin{bmatrix} a_{1,1} & a_{1,2} & a_{1,3} \\ a_{2,1} & a_{2,2} & a_{2,3} \end{bmatrix}$$

$$cA=c\begin{bmatrix} a_{1,1} & a_{1,2} & a_{1,3} \\ a_{2,1} & a_{2,2} & a_{2,3} \end{bmatrix}$$

$$cA=\begin{bmatrix} ca_{1,1} & ca_{1,2} & ca_{1,3} \\ ca_{2,1} & ca_{2,2} & ca_{2,3} \end{bmatrix}$$

INSPIRING CREATIVE AND INNOVATIVE MINDS

UTM UNIVERSITI TEKNOLOGI MALAYSIA

www.utm.my

example

$$A=\begin{bmatrix}3 & 2 & 0\\ 5 & 1 & -4\end{bmatrix} \qquad B=\begin{bmatrix}-1 & 4 & 2\\ 7 & -5 & 3\end{bmatrix}$$

$$2A=2\begin{bmatrix}3 & 2 & 0\\ 5 & 1 & -4\end{bmatrix}=\begin{bmatrix}6 & 4 & 0\\ 10 & 2 & -8\end{bmatrix}$$

$$3B=3\begin{bmatrix}-1 & 4 & 2\\ 7 & -5 & 3\end{bmatrix}=\begin{bmatrix}-3 & 12 & 6\\ 21 & -15 & 9\end{bmatrix}$$

UTM UNIVERSITI TEKNOLOGI MALAYSIA

www.utm.my

example

$$A=\begin{bmatrix}3 & 2 & 0\\ 5 & 1 & -4\end{bmatrix} \qquad B=\begin{bmatrix}-1 & 4 & 2\\ 7 & -5 & 3\end{bmatrix}$$

$$3B-2A=\begin{bmatrix}-3 & 12 & 6\\ 21 & -15 & 9\end{bmatrix}-\begin{bmatrix}6 & 4 & 0\\ 10 & 2 & -8\end{bmatrix}=\begin{bmatrix}-9 & 8 & 6\\ 11 & -17 & 17\end{bmatrix}$$

INSPIRING CREATIVE AND INNOVATIVE MINDS

UTM UNIVERSITI TEKNOLOGI MALAYSIA

www.utm.my

Theorem

- Let A and B denote matrices and let c and d denote numbers.
  - c(A+B) = cA+cB
  - (c+d)A = cA+dA
  - c(dA) = (cd)A
  - 1A = A

UTM UNIVERSITI TEKNOLOGI MALAYSIA

www.utm.my

Theorem

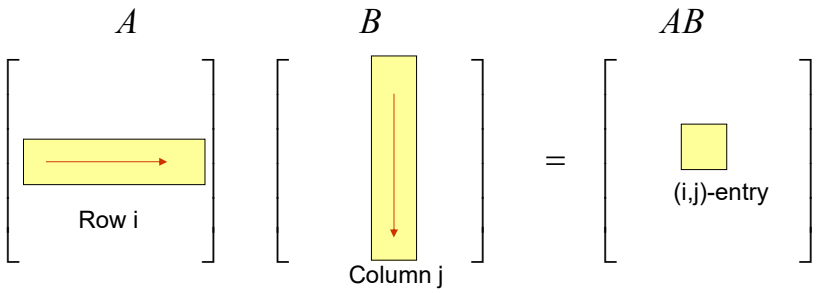
- If cA=0, then either c=0 or A=0
  - c(0) = 0
  - 0A = 0

UTM UNIVERSITI TEKNOLOGI MALAYSIA

www.utm.my

Matrix Multiplication

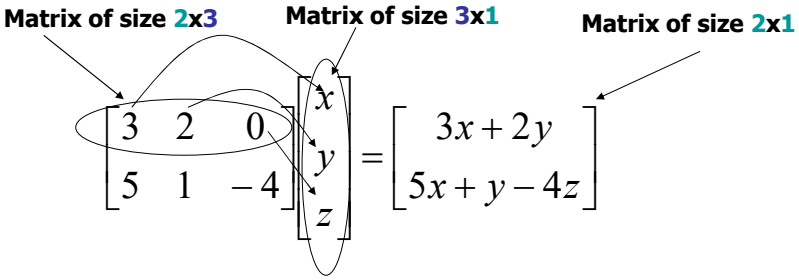
- The product of an m×n matrix A and an n×p matrix B is the m×p matrix whose (i,j)-entry is the dot product of row i of A and column j of B.



UTM UNIVERSITI TEKNOLOGI MALAYSIA

www.utm.my

Matrix Multiplication



INSPIRING CREATIVE AND INNOVATIVE MINDS

UTM UNIVERSITI TEKNOLOGI MALAYSIA

www.utm.my

Matrix Multiplication

$$\begin{bmatrix} 3 & 2 & 0 \\ 5 & 1 & -4 \end{bmatrix} \begin{bmatrix} x & a \\ y & b \\ z & c \end{bmatrix} = \begin{bmatrix} 3x+2y & 3a+2b \\ 5x+y-4z & 5a+b-4c \end{bmatrix}$$

Matrix of size 2x3 — Matrix of size 3x2 — Matrix of size 2x2

UTM UNIVERSITI TEKNOLOGI MALAYSIA

www.utm.my

example

$$\begin{bmatrix} 3 & 2 & 0 \\ 5 & 1 & -4 \end{bmatrix} \begin{bmatrix} 5 & 4 \\ 2 & 3 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 3.5+2.2+0.0 & 3.4+2.3+0.1 \\ 5.5+1.2+(-4).0 & 5.4+1.3+(-4).1 \end{bmatrix}$$

$$= \begin{bmatrix} 19 & 18 \\ 27 & 19 \end{bmatrix}$$

$$\begin{bmatrix} 3 & 2 & 0 \\ 5 & 1 & -4 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 3x+2y \\ 5x+y-4z \end{bmatrix}$$

INSPIRING CREATIVE AND INNOVATIVE MINDS

UTM UNIVERSITI TEKNOLOGI MALAYSIA

www.utm.my

Transposition

- Given an m×n matrix A, the transpose of A is the n×m matrix, denoted by A^T, whose columns are formed from the corresponding rows of A.

UTM UNIVERSITI TEKNOLOGI MALAYSIA

www.utm.my

example

$$A = \begin{bmatrix} 3 & 2 & 0 \\ 5 & 1 & -4 \end{bmatrix} \qquad B = \begin{bmatrix} w & x \\ y & z \end{bmatrix} \qquad C = \begin{bmatrix} 1 & 2 & 3 \end{bmatrix}$$

$$A^T = \begin{bmatrix} 3 & 5 \\ 2 & 1 \\ 0 & -4 \end{bmatrix} \qquad B^T = \begin{bmatrix} w & y \\ x & z \end{bmatrix} \qquad C^T = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$$

INSPIRING CREATIVE AND INNOVATIVE MINDS

UTM UNIVERSITI TEKNOLOGI MALAYSIA

www.utm.my

Theorem

- Let A and B denote matrices whose sizes are appropriate for the following sums and products.
  - $(A^T)^T = A$
  - $(A+B)^T = A^T + B^T$
  - For any scalar r, $(rA)^T = rA^T$
  - $(AB)^T = B^T A^T$

UTM UNIVERSITI TEKNOLOGI MALAYSIA

www.utm.my

Exercise

- Let

$$A = \begin{bmatrix} 2 & 1 & 0 \\ 5 & -1 & 3 \end{bmatrix} \quad B = \begin{bmatrix} 4 & 0 & -2 \\ 7 & 11 & 8 \end{bmatrix} \quad C = \begin{bmatrix} 2 & 1 \\ 0 & 4 \\ 1 & 3 \end{bmatrix}$$

- Compute:

(a) $A + B$ (b) $3A-B$ (c) AC (d) $A-2B$
(e) A^T+C (f) CB (g) A^T+B^T (h) C^T-B
(i) AA^T (j) A^TA (k) A^TB (l) C^T-A

INSPIRING CREATIVE AND INNOVATIVE MINDS



UTM
UNIVERSITI TEKNOLOGI MALAYSIA
www.utm.my

Determinants

- The determinant of a 2×2 matrix A= $\begin{bmatrix} a & b \\ c & d \end{bmatrix}$

$$\det A = \det \begin{bmatrix} a & b \\ c & d \end{bmatrix} = ad - bc$$

- For a 1×1 matrix $A=[a]$, det $A=a$.

UTM UNIVERSITI TEKNOLOGI MALAYSIA

www.utm.my

example

- Compute the determinant of $A = \begin{bmatrix} 2 & 1 \\ 5 & 4 \end{bmatrix}$

$$\det A = \det \begin{bmatrix} 2 & 1 \\ 5 & 4 \end{bmatrix} = 2.4 - 1.5 = 8 - 5 = 3$$

UTM UNIVERSITI TEKNOLOGI MALAYSIA

www.utm.my

Submatrix

- For any square matrix A, let A_{ij} denote the submatrix formed by deleting the i th row and j th column of A.
- Example:

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix} \qquad A_{12} = \begin{bmatrix} 4 & 6 \\ 7 & 9 \end{bmatrix}$$

INSPIRING CREATIVE AND INNOVATIVE MINDS

UTM UNIVERSITI TEKNOLOGI MALAYSIA

www.utm.my

Determinants (n ≥2)

- For n≥2, the determinant of an n×n matrix A=[a_{ij}] is the sum of n terms of the form $\pm a_{1j}$ det A_{1j}, with plus and minus signs alternating, where the entries a_{11}, a_{12},… , a_{1n} are from the first row of A.

$$\det A = a_{11} \det A_{11} - a_{12} \det A_{12} + \ldots + (-1)^{1+n} a_{1n} \det A_{1n}$$

UTM UNIVERSITI TEKNOLOGI MALAYSIA

www.utm.my

Cofactors

- Given A=[a_{ij}] the **(i.j)-cofactor** of A is the number C_{ij} given by

$$C_{ij} = (-1)^{i+j} \det A_{ij}$$

- Then

$$\det A = a_{11}C_{11} + a_{12}C_{12} + \ldots + a_{1n}C_{1n}$$

INSPIRING CREATIVE AND INNOVATIVE MINDS

UTM UNIVERSITI TEKNOLOGI MALAYSIA

www.utm.my

Determinants

- The + or – sign in the (i,j)-cofactor depends on the position of a_{ij} in the matrix.

$$\begin{bmatrix} + & - & + & - & \dots \\ - & + & - & + & \dots \\ + & - & + & - & \dots \\ - & + & - & + & \dots \\ \vdots & \vdots & \vdots & \vdots & \vdots \end{bmatrix}$$

UTM UNIVERSITI TEKNOLOGI MALAYSIA

www.utm.my

Determinants

- The determinant of an n×n matrix A can be computed by a cofactor expansion across any row or down any column.

INSPIRING CREATIVE AND INNOVATIVE MINDS

UTM UNIVERSITI TEKNOLOGI MALAYSIA

www.utm.my

Determinants

- The expansion across the i th row using the cofactor

$$\det A = a_{i1}C_{i1} + a_{i2}C_{i2} + \ldots + a_{in}C_{in}$$

- The cofactor expansion down the j th column is

$$\det A = a_{1j}C_{1j} + a_{2j}C_{2j} + \ldots + a_{nj}C_{nj}$$

UTM UNIVERSITI TEKNOLOGI MALAYSIA

www.utm.my

Determinants

- The determinant of a 3x3 matrix using first row expansion:

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$$

$$\det A = a_{11} \det \begin{bmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{bmatrix} - a_{12} \det \begin{bmatrix} a_{21} & a_{23} \\ a_{31} & a_{33} \end{bmatrix} + a_{13} \det \begin{bmatrix} a_{21} & a_{22} \\ a_{31} & a_{32} \end{bmatrix}$$

INSPIRING CREATIVE AND INNOVATIVE MINDS

UTM UNIVERSITI TEKNOLOGI MALAYSIA

www.utm.my

example

- Use a cofactor expansion across the 3rd row to compute det A.

$$A = \begin{bmatrix} 1 & 5 & 0 \\ 2 & 4 & -1 \\ 0 & -2 & 0 \end{bmatrix}$$

UTM UNIVERSITI TEKNOLOGI MALAYSIA

www.utm.my

example

$$A = \begin{bmatrix} 1 & 5 & 0 \\ 2 & 4 & -1 \\ 0 & -2 & 0 \end{bmatrix}$$

$$\det A = a_{31}C_{31} + a_{32}C_{32} + a_{33}C_{33}$$

$$= (-1)^{3+1} a_{31} \det A_{31} + (-1)^{3+2} a_{32} \det A_{32} + (-1)^{3+3} a_{33} \det A_{33}$$

$$= 0\begin{vmatrix} 5 & 0 \\ 4 & -1 \end{vmatrix} - (-2)\begin{vmatrix} 1 & 0 \\ 2 & -1 \end{vmatrix} + 0\begin{vmatrix} 1 & 5 \\ 2 & 4 \end{vmatrix}$$

$$= 0 + 2(-1) + 0 = -2$$

INSPIRING CREATIVE AND INNOVATIVE MINDS

UTM UNIVERSITI TEKNOLOGI MALAYSIA

www.utm.my

exercise

(a)

$$A = \begin{bmatrix} 3 & 0 & 4 \\ 2 & 3 & 2 \\ 0 & 5 & -1 \end{bmatrix}$$

(b)

$$A = \begin{bmatrix} 1 & 3 & 5 \\ 2 & 1 & 1 \\ 3 & 4 & 2 \end{bmatrix}$$

Compute the determinants of the matrices using a cofactor expansion

i. across the 1st row
ii. down the 2nd column

UTM UNIVERSITI TEKNOLOGI MALAYSIA

www.utm.my

Matrix Inverses

- If A is a square matrix, a matrix C is called an inverse of A if,

  AC= I and CA=I

- $C=A^{-1}$

INSPIRING CREATIVE AND INNOVATIVE MINDS

UTM UNIVERSITI TEKNOLOGI MALAYSIA

www.utm.my

Matrix Inverses

- If A = $\begin{bmatrix} a & b \\ c & d \end{bmatrix}$ and ad–bc ≠ 0,

$$A^{-1} = \frac{1}{ad-bc}\begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

- (ad–bc) is the determinant of A and $\begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$ is the adjoint of A.

UTM UNIVERSITI TEKNOLOGI MALAYSIA

www.utm.my

example

- Find the inverse of the matrix A= $\begin{bmatrix} 4 & 1 \\ 3 & 2 \end{bmatrix}$

$$A^{-1} = \frac{1}{4.2-1.3}\begin{bmatrix} 2 & -1 \\ -3 & 4 \end{bmatrix} = \begin{bmatrix} \frac{2}{5} & \frac{-1}{5} \\ \frac{-3}{5} & \frac{4}{5} \end{bmatrix}$$

$$\begin{bmatrix} 4 & 1 \\ 3 & 2 \end{bmatrix}\begin{bmatrix} \frac{2}{5} & \frac{-1}{5} \\ \frac{-3}{5} & \frac{4}{5} \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

UTM UNIVERSITI TEKNOLOGI MALAYSIA

www.utm.my

Matrix Inverses (3 x 3)

- Using cofactor matrix to compute the inverse:

$$A^{-1} = \frac{1}{|A|}(\text{adjoint } A) = \frac{1}{|A|}(\text{cofactormatrix } A)^T$$

UTM UNIVERSITI TEKNOLOGI MALAYSIA

www.utm.my

Example

$$A = \begin{pmatrix} 2 & 1 & 2 \\ -3 & -1 & -1 \\ 5 & 2 & 1 \end{pmatrix}$$

$$\text{Det}(A) = 2\begin{vmatrix} -1 & -1 \\ 2 & 1 \end{vmatrix} - 1\begin{vmatrix} -3 & -1 \\ 5 & 1 \end{vmatrix} + 2\begin{vmatrix} -3 & -1 \\ 5 & 2 \end{vmatrix}$$

$$= 2(-1+2) - 1(-3+5) + 2(-6+5)$$

$$= 2 - 2 - 2$$

$$= -2$$

INSPIRING CREATIVE AND INNOVATIVE MINDS

UTM UNIVERSITI TEKNOLOGI MALAYSIA

www.utm.my

Example(cont.)

$A_{11}= \begin{vmatrix} -1 & -1 \\ 2 & 1 \end{vmatrix} =$ (-1+2) =1

$A_{12} = \begin{vmatrix} -3 & -1 \\ 5 & 1 \end{vmatrix} = (-3+5) = 2$

$A_{13}= \begin{vmatrix} -3 & -1 \\ 5 & 2 \end{vmatrix} = (-6+5) = -1$

$A_{21}= \begin{vmatrix} 1 & 2 \\ 2 & 1 \end{vmatrix} = (1-4) = -3$

$A_{22}= \begin{vmatrix} 2 & 2 \\ 5 & 1 \end{vmatrix} = (2-10) = -8$

$A_{23}= \begin{vmatrix} 2 & 1 \\ 5 & 2 \end{vmatrix} = (4-5) = -1$

$A_{31}= \begin{vmatrix} 1 & 2 \\ -1 & -1 \end{vmatrix} = (-1+2) = 1$

$A_{32}= \begin{vmatrix} 2 & 2 \\ -3 & -1 \end{vmatrix} = (-2+6) = 4$

$A_{33}= \begin{vmatrix} 2 & 1 \\ -3 & -1 \end{vmatrix} = (-2+3) = 1$

INSPIRING CREATIVE AND INNOVATIVE MINDS

UTM UNIVERSITI TEKNOLOGI MALAYSIA

www.utm.my

Example(cont.)

Cofactor matrix A = $\begin{bmatrix} 1 & -2 & -1 \\ 3 & -8 & 1 \\ 1 & -4 & 1 \end{bmatrix}$

Ad joint A = $\begin{pmatrix} 1 & 3 & 1 \\ -2 & -8 & -4 \\ -1 & 1 & 1 \end{pmatrix}$

$A^{-1}= -\frac{1}{2}\begin{pmatrix} 1 & 3 & 1 \\ -2 & -8 & -4 \\ -1 & 1 & 1 \end{pmatrix}$

$= \begin{pmatrix} -\frac{1}{2} & \frac{3}{2} & -\frac{1}{2} \\ 1 & 4 & 2 \\ \frac{1}{2} & -\frac{1}{2} & -\frac{1}{2} \end{pmatrix}$

INSPIRING CREATIVE AND INNOVATIVE MINDS

UTM UNIVERSITI TEKNOLOGI MALAYSIA

www.utm.my

exercise

- Find the inverse of the given matrices.

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 0 & 4 & 5 \\ 1 & 0 & 6 \end{bmatrix} \qquad B = \begin{bmatrix} 3 & 5 \\ 1 & 2 \end{bmatrix}$$

UTM UNIVERSITI TEKNOLOGI MALAYSIA

www.utm.my

INSPIRING CREATIVE AND INNOVATIVE MINDS



UTM UNIVERSITI TEKNOLOGI MALAYSIA

www.utm.my

System of Linear Equations

- A finite collection of linear equations is called a system of linear equations.
- A system of *m* equations in *n* variables,

$$\begin{aligned} a_{11}x_1 + a_{12}x_2 + ... + a_{1n}x_n &= b_1 \\ a_{21}x_1 + a_{22}x_2 + ... + a_{2n}x_n &= b_2 \\ \vdots \qquad\quad \vdots \qquad\quad \vdots & \\ a_{m1}x_1 + a_{m2}x_2 + ... + a_{mn}x_n &= b_m \end{aligned}$$

UTM UNIVERSITI TEKNOLOGI MALAYSIA

www.utm.my

System of Linear Equations

- Matrix form, *AX=B*

$$\begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & & \vdots \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{bmatrix}$$

INSPIRING CREATIVE AND INNOVATIVE MINDS

UTM UNIVERSITI TEKNOLOGI MALAYSIA

www.utm.my

System of Linear Equations

- *A* is the matrix of coefficients.
- *X* is the matrix of variables.
- *B* is the matrix of constants.

UTM UNIVERSITI TEKNOLOGI MALAYSIA

www.utm.my

System of Linear Equations

- Augmented matrix, [*A*|*B*]

$$\left[\begin{array}{cccc|c} a_{11} & a_{12} & \cdots & a_{1n} & b_1 \\ a_{21} & a_{22} & \cdots & a_{2n} & b_2 \\ \vdots & \vdots & & \vdots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} & b_m \end{array}\right]$$

UTM UNIVERSITI TEKNOLOGI MALAYSIA

www.utm.my

example

- A system,

$$x_1 + x_2 + 5x_3 = 5$$
$$3x_1 + 2x_2 - x_3 = 10$$
$$2x_1 - 4x_2 + x_3 = 8$$

$AX=B$

$$\begin{bmatrix} 1 & 1 & 5 \\ 3 & 2 & -1 \\ 2 & -4 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 5 \\ 10 \\ 8 \end{bmatrix}$$

$[A|B]$

$$\left[\begin{array}{ccc|c} 1 & 1 & 5 & 5 \\ 3 & 2 & -1 & 10 \\ 2 & -4 & 1 & 8 \end{array}\right]$$

UTM UNIVERSITI TEKNOLOGI MALAYSIA

www.utm.my

Row-echelon Matrices

- A matrix is said to be row-echelon form if the following conditions are satisfied:
  - All zero rows (if any) are at the bottom of the matrix.
  - If two successive rows are non zero, the second row starts with more zero than the first (moving from left to right)

INSPIRING CREATIVE AND INNOVATIVE MINDS

- Every row with all 0 entries is below every row with nonzero entries
- Each leading entry of a row is in a column to the right of the leading entry of the row above it(leading entry-any nonzero value)

(The second row starts with more zero than the first(left to right)

- All entries in a column below a leading entry are zeros

$$\begin{pmatrix} \# & * & * & * & * & * \\ 0 & 0 & \# & * & * & * \\ 0 & 0 & 0 & 0 & 0 & \# \\ 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix}$$

#-leading entry(left most non zero entry)
***-any number/values including 0**

UTM UNIVERSITI TEKNOLOGI MALAYSIA
www.utm.my

Reduced Row-echelon Matrices

- A row-echelon matrix is said to be in reduced row-echelon form if, in addition, it satisfies:
  - The leading (leftmost non zero) entry in each non zero row is 1 (called the leading 1)
  - Each leading 1 is the only nonzero entry in its column.

INSPIRING CREATIVE AND INNOVATIVE MINDS

UTM UNIVERSITI TEKNOLOGI MALAYSIA

www.utm.my

example

- The matrices

$$\begin{bmatrix} 2 & 1 & 4 & 0 & 1 \\ 0 & 0 & 1 & 0 & 2 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} \qquad \begin{bmatrix} 3 & 2 & 1 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$

are in row-echelon form.

UTM UNIVERSITI TEKNOLOGI MALAYSIA

www.utm.my

example

- The matrices

$$\begin{bmatrix} 1 & 3 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 & 2 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} \qquad \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$

are in reduced row-echelon form

UTM UNIVERSITI TEKNOLOGI MALAYSIA

www.utm.my

example

- The matrices

$$\begin{bmatrix} 3 & 2 & 1 \\ 0 & 2 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \qquad \begin{bmatrix} 2 & 1 & 4 & 0 & 1 \\ 0 & 0 & 1 & 2 & 0 \\ 0 & 0 & 1 & 0 & 1 \end{bmatrix}$$

are not in reduced row-echelon form or row-echelon form

UTM UNIVERSITI TEKNOLOGI MALAYSIA

www.utm.my

Elementary Row Operation

- Elementary row operations can be used to transform the matrix into its row-echelon form.
- If we denote row 1 by R1, row 2 by R2, etc., and a is a scalar, the three elementary row operations are as follows:
  1) Swap two rows (equations), denoted R1↔R2 (this would swap rows 1 and 2)
  2) Multiply a row by a nonzero scalar, denoted a R1 (this would multiply row 1 by a)
  3) Add a multiple of a row to another row, denoted R1 + a R2 (this would add a time row 2 to row 1, and replace the previous row 1)

INSPIRING CREATIVE AND INNOVATIVE MINDS

UTM UNIVERSITI TEKNOLOGI MALAYSIA

www.utm.my

Example

- Given is a linear equation system,

$$2x_1 + x_2 + x_3 = 7$$
$$3x_1 + 2x_2 - x_3 = 4$$
$$x_1 - 4x_2 + 2x_2 = -1$$

- Augmented form

$$\left(\begin{array}{ccc|c} 2 & 1 & 1 & 7 \\ 3 & 2 & -1 & 4 \\ 1 & -4 & 2 & -1 \end{array}\right)$$

UTM UNIVERSITI TEKNOLOGI MALAYSIA

www.utm.my

- Swap two rows

$$\left(\begin{array}{ccc|c} 2 & 1 & 1 & 7 \\ 3 & 2 & -1 & 4 \\ 1 & -4 & 2 & -1 \end{array}\right) \quad R_1 \longleftrightarrow R_3 \quad \left(\begin{array}{ccc|c} 1 & -4 & 2 & -1 \\ 3 & 2 & -1 & 4 \\ 2 & 1 & 1 & 7 \end{array}\right)$$

INSPIRING CREATIVE AND INNOVATIVE MINDS

- Multiply row 1 with a =3, *3R1* and add the multiple row to another row, *R2*

$$\left(\begin{array}{ccc|c} 1 & -4 & 2 & -1 \\ 3 & 2 & -1 & 4 \\ 2 & 1 & 1 & 7 \end{array}\right) \longrightarrow R_2 \to R_2 - 3R_1 \left(\begin{array}{ccc|c} 1 & -4 & 2 & -1 \\ 0 & 14 & -7 & 7 \\ 2 & 1 & 1 & 7 \end{array}\right)$$

$R_2 \to R_2 - 3R_1 =$

$$3 - 3(1) = 0$$
$$2 - 3(-4) = 14$$
$$-1 - 3(2) = -7$$
$$4 - 3(-1) = 7$$

The notation means to take row 2 and subtract 3 times row 1 from it to produce the new augmented matrix.

Gaussian Elimination

- Assume that a system of linear equations has one solution.
  - Carry the augmented matrix of the system to row-echelon form.
  - Assign the nonleading variables as parameters.
  - Solve the leading variables using back-substitution.

UTM UNIVERSITI TEKNOLOGI MALAYSIA
www.utm.my

Gaussian Elimination

- Every matrix can be carried to row-echelon form by a sequence of elementary row operations:
  - Interchanging two rows.
  - Multiplying a row by a non zero scalar.
  - Adding a multiple of one row to another row.

INSPIRING CREATIVE AND INNOVATIVE MINDS

UTM UNIVERSITI TEKNOLOGI MALAYSIA
www.utm.my

example

- Solve the system of linear equations

$$2x_1 + x_2 + x_3 = 7$$
$$3x_1 + 2x_2 - x_3 = 4$$
$$x_1 - 4x_2 + 2x_3 = -1$$

INSPIRING CREATIVE AND INNOVATIVE MINDS

UTM UNIVERSITI TEKNOLOGI MALAYSIA

www.utm.my

example

- The augmented matrix for this system

$$\begin{aligned} 2x_1 + x_2 + x_3 &= 7 \\ 3x_1 + 2x_2 - x_3 &= 4 \\ x_1 - 4x_2 + 2x_3 &= -1 \end{aligned} \quad \rightarrow \quad \left[\begin{array}{ccc|c} 2 & 1 & 1 & 7 \\ 3 & 2 & -1 & 4 \\ 1 & -4 & 2 & -1 \end{array}\right]$$

UTM UNIVERSITI TEKNOLOGI MALAYSIA

www.utm.my

example

- Reduce the augmented matrix by elementary row operations to row echelon form

$$\left[\begin{array}{ccc|c} 2 & 1 & 1 & 7 \\ 3 & 2 & -1 & 4 \\ 1 & -4 & 2 & -1 \end{array}\right] \quad B_1 \leftrightarrow B_3 \rightarrow \quad \left[\begin{array}{ccc|c} 1 & -4 & 2 & -1 \\ 3 & 2 & -1 & 4 \\ 2 & 1 & 1 & 7 \end{array}\right]$$

UTM UNIVERSITI TEKNOLOGI MALAYSIA

www.utm.my

example

$$m_{21} = -\frac{a_{21}}{a_{11}} = -\frac{3}{1} = -3, \qquad m_{31} = -\frac{a_{31}}{a_{11}} = -\frac{2}{1} = -2$$

$$\left[\begin{array}{ccc|c} 1 & -4 & 2 & -1 \\ 3 & 2 & -1 & 4 \\ 2 & 1 & 1 & 7 \end{array}\right]$$

$$B_2 \to B_2 + m_{21}B_1$$
$$B_3 \to B_3 + m_{31}B_1$$

3+(-3)(1)=0
2+(-3)(-4)=14
-1+(-3)(2)=-7
4+(-3)(-1)=7

2+(-2)(1)=0
1+(-2)(-4)=9
1+(-2)(2)=-3
7+(-2)(-1)=9

$$\left[\begin{array}{ccc|c} 1 & -4 & 2 & -1 \\ 0 & 14 & -7 & 7 \\ 0 & 9 & -3 & 9 \end{array}\right]$$

UTM UNIVERSITI TEKNOLOGI MALAYSIA

www.utm.my

example

$$\left[\begin{array}{ccc|c} 1 & -4 & 2 & -1 \\ 0 & 14 & -7 & 7 \\ 0 & 9 & -3 & 9 \end{array}\right] \qquad B_2 \to (\tfrac{1}{7})B_2 \qquad \left[\begin{array}{ccc|c} 1 & -4 & 2 & -1 \\ 0 & 2 & -1 & 1 \\ 0 & 9 & -3 & 9 \end{array}\right]$$

INSPIRING CREATIVE AND INNOVATIVE MINDS

UTM UNIVERSITI TEKNOLOGI MALAYSIA

www.utm.my

example

$$m_{32} = -\frac{a_{32}}{a_{22}} = -\frac{9}{2}$$

0+(-9/2)(0)=0
9+(-9/2)(2)=0
-3+(-9/2)(-1)=3/2
9+(-9/2)(1)=9/2

$$\left[\begin{array}{ccc|c} 1 & -4 & 2 & -1 \\ 0 & 2 & -1 & 1 \\ 0 & 9 & -3 & 9 \end{array}\right] \qquad B_3 \to B_3 + m_{32}B_2$$

$$\left[\begin{array}{ccc|c} 1 & -4 & 2 & -1 \\ 0 & 2 & -1 & 1 \\ 0 & 0 & \frac{3}{2} & \frac{9}{2} \end{array}\right]$$

UTM UNIVERSITI TEKNOLOGI MALAYSIA

www.utm.my

example

- Row echelon form

$$\left[\begin{array}{ccc|c} 1 & -4 & 2 & -1 \\ 0 & 2 & -1 & 1 \\ 0 & 0 & \frac{3}{2} & \frac{9}{2} \end{array}\right] \rightarrow \begin{aligned} x_1 - 4x_2 + 2x_3 &= -1 \\ 2x_2 - x_3 &= 1 \\ \tfrac{3}{2}x_3 &= \tfrac{9}{2} \end{aligned}$$

INSPIRING CREATIVE AND INNOVATIVE MINDS

UTM UNIVERSITI TEKNOLOGI MALAYSIA

www.utm.my

example

- Back-substitution

$$x_1 - 4x_2 + 2x_3 = -1$$
$$2x_2 - x_3 = 1$$
$$\tfrac{3}{2}x_3 = \tfrac{9}{2}$$

$$x_3 = 3$$

$$x_2 = \frac{1+3}{2} = 2$$

$$x_1 = -1 + 4(2) - 2(3) = 1$$

UTM UNIVERSITI TEKNOLOGI MALAYSIA

www.utm.my

example

- Solve the system of linear equations

$$x_1 + 2x_2 + 3x_3 + 4x_4 = 10$$
$$x_1 + 2x_2 + 4x_3 + 5x_4 = 8$$
$$2x_1 + 4x_2 + 6x_3 + 8x_4 = 20$$

INSPIRING CREATIVE AND INNOVATIVE MINDS

UTM UNIVERSITI TEKNOLOGI MALAYSIA

www.utm.my

example

- The augmented matrix for this system

$$\begin{aligned} x_1 + 2x_2 + 3x_3 + 4x_4 &= 10 \\ x_1 + 2x_2 + 4x_3 + 5x_4 &= 8 \\ 2x_1 + 4x_2 + 6x_3 + 8x_4 &= 20 \end{aligned} \quad \rightarrow \quad \left[\begin{array}{cccc|c} 1 & 2 & 3 & 4 & 10 \\ 1 & 2 & 4 & 5 & 8 \\ 2 & 4 & 6 & 8 & 20 \end{array}\right]$$

UTM UNIVERSITI TEKNOLOGI MALAYSIA

www.utm.my

example

$$m_{21} = -\frac{a_{21}}{a_{11}} = -\frac{1}{1} = -1$$

1+(-1)(1)=0
2+(-1)(2)=0
4+(-1)(3)=1
5+(-1)(4)=1
8+(-1)(10)=-2

$$\left[\begin{array}{cccc|c} 1 & 2 & 3 & 4 & 10 \\ 1 & 2 & 4 & 5 & 8 \\ 2 & 4 & 6 & 8 & 20 \end{array}\right] \qquad B_2 \rightarrow B_2 + m_{21}B_1$$

$$\left[\begin{array}{cccc|c} 1 & 2 & 3 & 4 & 10 \\ 0 & 0 & 1 & 1 & -2 \\ 2 & 4 & 6 & 8 & 20 \end{array}\right]$$

INSPIRING CREATIVE AND INNOVATIVE MINDS

UTM UNIVERSITI TEKNOLOGI MALAYSIA

www.utm.my

example

$$m_{31} = -\frac{a_{31}}{a_{11}} = -\frac{2}{1} = -2$$

$$\left[\begin{array}{cccc|c} 1 & 2 & 3 & 4 & 10 \\ 0 & 0 & 1 & 1 & -2 \\ 2 & 4 & 6 & 8 & 20 \end{array}\right] \qquad B_3 \to B_3 + m_{31}B_1$$

2+(-2)(1)=0
4+(-2)(2)=0
6+(-2)(3)=0
8+(-2)(4)=0
20+(-2)(10)=0

$$\left[\begin{array}{cccc|c} 1 & 2 & 3 & 4 & 10 \\ 0 & 0 & 1 & 1 & -2 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right]$$

UTM UNIVERSITI TEKNOLOGI MALAYSIA

www.utm.my

example

$$\left[\begin{array}{cccc|c} 1 & 2 & 3 & 4 & 10 \\ 0 & 0 & 1 & 1 & -2 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right]$$

Let nonleading variables

$x_2 = s$ and $x_4 = t$

(assign nonleading variables as parameter)

leading variables,

(solve leading variables using back-substitutions)

$$x_3 = -2 - t$$

$$x_1 = 10 - 2s - 3(-2 - t) - 4t = 16 - 2s - t$$

INSPIRING CREATIVE AND INNOVATIVE MINDS

UTM UNIVERSITI TEKNOLOGI MALAYSIA

www.utm.my

exercise

- Consider the following systems of linear equations

(a)
$$\begin{aligned} 2x_1 + x_2 + x_3 &= 7 \\ 3x_1 + 2x_2 - x_3 &= 4 \\ x_1 - 4x_2 + 2x_3 &= -1 \end{aligned}$$

(b)
$$\begin{aligned} x_1 - 3x_2 + x_3 + x_4 - x_5 &= 8 \\ -2x_1 + 6x_2 + x_3 - 2x_4 - 4x_5 &= -1 \\ 3x_1 - 9x_2 + 8x_3 + 4x_4 - 13x_5 &= 49 \end{aligned}$$

- Use Gaussian Elimination to solve the systems.



UTM UNIVERSITI TEKNOLOGI MALAYSIA

www.utm.my

INSPIRING CREATIVE AND INNOVATIVE MINDS



PART E

UTM UNIVERSITI TEKNOLOGI MALAYSIA

www.utm.my

Gauss-Jordan Elimination

- In Gauss-Jordan elimination, elements above the diagonal are eliminated in the same manner as are elements below the diagonal, thus avoiding the back substitution phase of the solution process.
- The elimination step results in an identity matrix rather than a triangular matrix.

UTM UNIVERSITI TEKNOLOGI MALAYSIA

www.utm.my

Gauss-Jordan Elimination

$$\left[\begin{array}{ccc|c} a_{11} & a_{12} & a_{13} & b_1 \\ a_{21} & a_{22} & a_{23} & b_2 \\ a_{31} & a_{32} & a_{33} & b_3 \end{array}\right] \rightarrow \left[\begin{array}{ccc|c} u_{11} & u_{12} & u_{13} & d_1 \\ 0 & u_{22} & u_{23} & d_2 \\ 0 & 0 & u_{33} & d_3 \end{array}\right]$$

$$\left[\begin{array}{ccc|c} u_{11} & u_{12} & u_{13} & d_1 \\ 0 & u_{22} & u_{23} & d_2 \\ 0 & 0 & u_{33} & d_3 \end{array}\right] \rightarrow \left[\begin{array}{ccc|c} d_{11} & 0 & 0 & e_1 \\ 0 & d_{22} & 0 & e_2 \\ 0 & 0 & d_{33} & e_3 \end{array}\right]$$

INSPIRING CREATIVE AND INNOVATIVE MINDS

UTM UNIVERSITI TEKNOLOGI MALAYSIA

www.utm.my

Gauss-Jordan Elimination

$$\left[\begin{array}{ccc|c} d_{11} & 0 & 0 & e_1 \\ 0 & d_{22} & 0 & e_2 \\ 0 & 0 & d_{33} & e_3 \end{array}\right] \rightarrow \begin{bmatrix} d_{11} & 0 & 0 \\ 0 & d_{22} & 0 \\ 0 & 0 & d_{33} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} e_1 \\ e_2 \\ e_3 \end{bmatrix}$$

$$\begin{aligned} d_{11}x_1 &= e_1 \\ d_{22}x_2 &= e_2 \\ d_{33}x_3 &= e_3 \end{aligned} \qquad x_1 = \frac{e_1}{d_{11}}, \quad x_2 = \frac{e_2}{d_{22}}, \quad x_3 = \frac{e_3}{d_{33}}$$

UTM UNIVERSITI TEKNOLOGI MALAYSIA

www.utm.my

Example

- Refer to previous example (Gauss Elimination)

$$\left[\begin{array}{ccc|c} 1 & -2 & 4 & -2 \\ 0 & 1 & 5 & -1 \\ 0 & 4 & -11 & 9 \end{array}\right] \rightarrow \left[\begin{array}{ccc|c} 1 & -2 & 4 & -2 \\ 0 & 1 & 5 & -1 \\ 0 & 0 & -31 & 13 \end{array}\right]$$

$\mathbf{R_3 \rightarrow R_3 - 4R_2}$

INSPIRING CREATIVE AND INNOVATIVE MINDS

UTM UNIVERSITI TEKNOLOGI MALAYSIA

www.utm.my

Example

- Column 2, row 1

$$m_{12} = -\frac{a_{12}}{a_{22}} = -\frac{-2}{1} = 2$$

R_1	$+ m_{12}R_2$	$= R_1$
1	+ (2)(0)	= 1
–2	+ (2)(1)	= 0
4	+ (2)(5)	= 14
–2	+ (2)(– 1)	= –4

$$\left[\begin{array}{ccc|c} 1 & -2 & 4 & -2 \\ 0 & 1 & 5 & -1 \\ 0 & 0 & -31 & 13 \end{array}\right]$$

UTM UNIVERSITI TEKNOLOGI MALAYSIA

www.utm.my

Example

$$\left[\begin{array}{ccc|c} 1 & -2 & 4 & -2 \\ 0 & 1 & 5 & -1 \\ 0 & 0 & -31 & 13 \end{array}\right] \rightarrow \left[\begin{array}{ccc|c} 1 & 0 & 14 & -4 \\ 0 & 1 & 5 & -1 \\ 0 & 0 & -31 & 13 \end{array}\right]$$

INSPIRING CREATIVE AND INNOVATIVE MINDS

UTM UNIVERSITI TEKNOLOGI MALAYSIA

www.utm.my

Example

- Column 3, row 1

$$m_{13} = -\frac{a_{13}}{a_{33}} = -\frac{14}{-31} = \frac{14}{31}$$

R_1	$+ m_{13}R_3$	$= R_1$
1	+ (14/31)(0)	= 1
0	+ (14/31)(0)	= 0
14	+ (14/31)(– 31)	= 0
–4	+ (14/31)(13)	= 58/31

$$\left[\begin{array}{ccc|c} 1 & 0 & 14 & -4 \\ 0 & 1 & 5 & -1 \\ 0 & 0 & -31 & 13 \end{array}\right]$$

UTM UNIVERSITI TEKNOLOGI MALAYSIA

www.utm.my

Example

- Column 3 * 1/-31

$$\left[\begin{array}{ccc|c} 1 & 0 & 0 & 58/31 \\ 0 & 1 & 5 & -1 \\ 0 & 0 & 1 & 13/-31 \end{array}\right]$$

INSPIRING CREATIVE AND INNOVATIVE MINDS

UTM UNIVERSITI TEKNOLOGI MALAYSIA

www.utm.my

Example

- Column 3, row 2

$$m_{23} = -\frac{a_{23}}{a_{33}} = -\frac{5}{-1} = -5$$

R_2	$+ m_{23}R_3$	$= R_2$
0	+ (-5)(0)	= 0
1	+ (-5)(0)	= 1
5	+ (-5)(1)	= 0
–1	+ (-5)(13/-31)	= 34/31

$$\left[\begin{array}{ccc|c} 1 & 0 & 0 & 58/31 \\ 0 & 1 & 5 & -1 \\ 0 & 0 & 1 & 13/-31 \end{array}\right]$$

UTM UNIVERSITI TEKNOLOGI MALAYSIA

www.utm.my

Example

$$\left[\begin{array}{ccc|c} 1 & 0 & 0 & \frac{58}{31} \\ 0 & 1 & 0 & \frac{34}{31} \\ 0 & 0 & 1 & 13 \end{array}\right]$$

$$x = \left[\frac{58}{31}, \frac{34}{31}, -\frac{13}{31}\right]^T$$

INSPIRING CREATIVE AND INNOVATIVE MINDS

UTM UNIVERSITI TEKNOLOGI MALAYSIA

www.utm.my

Gauss-Jordan Elimination

- Although the Gauss-Jordan technique and Gauss elimination might appear almost identical, the former requires more work.
- Gauss-Jordan involves approximately 50 percent more operations than Gauss elimination.
- Therefore, Gauss elimination is the simple elimination method of preference for obtaining solutions of linear equations.

UTM UNIVERSITI TEKNOLOGI MALAYSIA

www.utm.my

Exercise

- Solve the following linear system using Gauss-Jordan elimination.

$$2x_1 + 4x_2 - 6x_3 = -4$$
$$x_1 + 5x_2 + 3x_3 = 10$$
$$x_1 + 3x_2 + 2x_3 = 5$$

Use fraction in your calculations.

Check your answers by substituting them into the original equations.



UTM
UNIVERSITI TEKNOLOGI MALAYSIA
www.utm.my

Matrix Factorization

- The solution of a system of linear equations $AX=B$ can be computed much more quickly if the matrix A can be factored in the form

  $A=LU$

  where

  L is a lower triangular matrix

  U is an upper triangular matrix

UTM UNIVERSITI TEKNOLOGI MALAYSIA

www.utm.my

Matrix Factorization

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix} \qquad (Ax=B)$$

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} = \begin{bmatrix} l_{11} & 0 & 0 \\ l_{21} & l_{22} & 0 \\ l_{31} & l_{32} & l_{33} \end{bmatrix} \begin{bmatrix} u_{11} & u_{12} & u_{13} \\ 0 & u_{22} & u_{23} \\ 0 & 0 & u_{33} \end{bmatrix} \qquad (A=LU)$$

UTM UNIVERSITI TEKNOLOGI MALAYSIA

www.utm.my

Matrix Factorization

$$\begin{bmatrix} l_{11} & 0 & 0 \\ l_{21} & l_{22} & 0 \\ l_{31} & l_{32} & l_{33} \end{bmatrix} \begin{bmatrix} u_{11} & u_{12} & u_{13} \\ 0 & u_{22} & u_{23} \\ 0 & 0 & u_{33} \end{bmatrix} = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$$

$$\begin{bmatrix} l_{11}u_{11} & l_{11}u_{12} & l_{11}u_{13} \\ l_{21}u_{11} & l_{21}u_{12}+l_{22}u_{22} & l_{21}u_{13}+l_{22}u_{23} \\ l_{31}u_{11} & l_{31}u_{12}+l_{32}u_{22} & l_{31}u_{13}+l_{32}u_{23}+l_{33}u_{33} \end{bmatrix} = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$$

INSPIRING CREATIVE AND INNOVATIVE MINDS

UTM UNIVERSITI TEKNOLOGI MALAYSIA

www.utm.my

Matrix Factorization

- *AX=B* can be solved in 2 stages:
  - First solve *LY=B* for *Y* by forward substitution.
  - Then solve *UX=Y* for *X* by back substitution.

UTM UNIVERSITI TEKNOLOGI MALAYSIA

www.utm.my

Matrix Factorization

- LY=B

$$\begin{bmatrix} l_{11} & 0 & 0 \\ l_{21} & l_{22} & 0 \\ l_{31} & l_{32} & l_{33} \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}$$

- Forward substitution

$$y_1 = \frac{b_1}{l_{11}},$$

$$y_2 = \frac{b_2 - l_{21} y_1}{l_{22}},$$

$$y_3 = \frac{b_3 - l_{31} y_1 - l_{32} y_2}{l_{33}}$$

INSPIRING CREATIVE AND INNOVATIVE MINDS

UTM UNIVERSITI TEKNOLOGI MALAYSIA

www.utm.my

Matrix Factorization

- UX=Y

$$\begin{bmatrix} u_{11} & u_{12} & u_{13} \\ 0 & u_{22} & u_{23} \\ 0 & 0 & u_{33} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix}$$

- Back substitution

$$x_3 = \frac{y_3}{u_{33}},$$

$$x_2 = \frac{y_2 - u_{23}x_3}{u_{22}},$$

$$x_1 = \frac{y_1 - u_{12}x_2 - u_{13}x_3}{u_{11}}$$

UTM UNIVERSITI TEKNOLOGI MALAYSIA

www.utm.my

Doolittle Factorization

- The diagonal elements of matrix *L* are 1's.

$$\underset{A}{\begin{bmatrix} a_{11} & a_{12} & .. & a_{1n} \\ a_{21} & a_{22} & .. & a_{2n} \\ : & : & .. & : \\ a_{n1} & a_{n2} & .. & a_{nn} \end{bmatrix}} = \underset{L}{\begin{bmatrix} 1 & 0 & .. & 0 \\ l_{21} & 1 & .. & 0 \\ : & : & .. & : \\ l_{n1} & l_{n2} & .. & 1 \end{bmatrix}} \underset{U}{\begin{bmatrix} u_{11} & u_{12} & .. & u_{1n} \\ 0 & u_{22} & .. & u_{2n} \\ : & : & .. & : \\ 0 & 0 & .. & u_{nn} \end{bmatrix}}$$

INSPIRING CREATIVE AND INNOVATIVE MINDS

UTM UNIVERSITI TEKNOLOGI MALAYSIA

www.utm.my

Doolittle Factorization

$$\begin{bmatrix} 1 & 0 & .. & 0 \\ l_{21} & 1 & .. & 0 \\ : & : & .. & : \\ l_{n1} & l_{n2} & .. & 1 \end{bmatrix} \begin{bmatrix} u_{11} & u_{12} & .. & u_{1n} \\ 0 & u_{22} & .. & u_{2n} \\ : & : & .. & : \\ 0 & 0 & .. & u_{nn} \end{bmatrix} = \begin{bmatrix} a_{11} & a_{12} & .. & a_{1n} \\ a_{21} & a_{22} & .. & a_{2n} \\ : & : & .. & : \\ a_{n1} & a_{n2} & .. & a_{nn} \end{bmatrix}$$

UTM UNIVERSITI TEKNOLOGI MALAYSIA

www.utm.my

example

- Find the *LU* factorization of the matrix using Doolittle form and use it to solve the linear system.

$$\begin{bmatrix} 1 & 2 & 1 \\ 3 & 1 & 1 \\ 1 & 4 & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 8 \\ 11 \\ 14 \end{bmatrix}$$

INSPIRING CREATIVE AND INNOVATIVE MINDS

UTM UNIVERSITI TEKNOLOGI MALAYSIA

www.utm.my

example

$$\begin{bmatrix} 1 & 0 & 0 \\ l_{21} & 1 & 0 \\ l_{31} & l_{32} & 1 \end{bmatrix} \begin{bmatrix} u_{11} & u_{12} & u_{13} \\ 0 & u_{22} & u_{23} \\ 0 & 0 & u_{33} \end{bmatrix} = \begin{bmatrix} 1 & 2 & 1 \\ 3 & 1 & 1 \\ 1 & 4 & 2 \end{bmatrix}$$

$$u_{11} = 1 \qquad u_{12} = 2 \qquad u_{13} = 1$$

UTM UNIVERSITI TEKNOLOGI MALAYSIA

www.utm.my

example

$$\begin{bmatrix} 1 & 0 & 0 \\ l_{21} & 1 & 0 \\ l_{31} & l_{32} & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 & 1 \\ 0 & u_{22} & u_{23} \\ 0 & 0 & u_{33} \end{bmatrix} = \begin{bmatrix} 1 & 2 & 1 \\ 3 & 1 & 1 \\ 1 & 4 & 2 \end{bmatrix}$$

$$l_{21} = \frac{3}{1} = 3 \qquad l_{31} = \frac{1}{1} = 1$$

INSPIRING CREATIVE AND INNOVATIVE MINDS

UTM UNIVERSITI TEKNOLOGI MALAYSIA

www.utm.my

example

$$\begin{bmatrix} 1 & 0 & 0 \\ 3 & 1 & 0 \\ 1 & l_{32} & 1 \end{bmatrix}\begin{bmatrix} 1 & 2 & 1 \\ 0 & u_{22} & u_{23} \\ 0 & 0 & u_{33} \end{bmatrix} = \begin{bmatrix} 1 & 2 & 1 \\ 3 & 1 & 1 \\ 1 & 4 & 2 \end{bmatrix}$$

$$u_{22} = 1-(3)(2) = -5 \qquad u_{23} = 1-(3)(1) = -2$$

UTM UNIVERSITI TEKNOLOGI MALAYSIA

www.utm.my

example

$$\begin{bmatrix} 1 & 0 & 0 \\ 3 & 1 & 0 \\ 1 & l_{32} & 1 \end{bmatrix}\begin{bmatrix} 1 & 2 & 1 \\ 0 & -5 & -2 \\ 0 & 0 & u_{33} \end{bmatrix} = \begin{bmatrix} 1 & 2 & 1 \\ 3 & 1 & 1 \\ 1 & 4 & 2 \end{bmatrix}$$

$$l_{32} = \frac{4-(1)(2)}{-5} = -\frac{2}{5}$$

INSPIRING CREATIVE AND INNOVATIVE MINDS

UTM UNIVERSITI TEKNOLOGI MALAYSIA

www.utm.my

example

$$\begin{bmatrix} 1 & 0 & 0 \\ 3 & 1 & 0 \\ 1 & \frac{-2}{5} & 1 \end{bmatrix}\begin{bmatrix} 1 & 2 & 1 \\ 0 & -5 & -2 \\ 0 & 0 & u_{33} \end{bmatrix} = \begin{bmatrix} 1 & 2 & 1 \\ 3 & 1 & 1 \\ 1 & 4 & 2 \end{bmatrix}$$

$$u_{33} = 2 - (1)(1) - \left(-\frac{2}{5}\right)(-2) = \frac{1}{5}$$

UTM UNIVERSITI TEKNOLOGI MALAYSIA

www.utm.my

example

$$\begin{bmatrix} 1 & 0 & 0 \\ 3 & 1 & 0 \\ 1 & \frac{-2}{5} & 1 \end{bmatrix}\begin{bmatrix} 1 & 2 & 1 \\ 0 & -5 & -2 \\ 0 & 0 & \frac{1}{5} \end{bmatrix} = \begin{bmatrix} 1 & 2 & 1 \\ 3 & 1 & 1 \\ 1 & 4 & 2 \end{bmatrix}$$

INSPIRING CREATIVE AND INNOVATIVE MINDS

UTM UNIVERSITI TEKNOLOGI MALAYSIA

www.utm.my

example

$$\begin{bmatrix} 1 & 0 & 0 \\ 3 & 1 & 0 \\ 1 & -\frac{2}{5} & 1 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix} = \begin{bmatrix} 8 \\ 11 \\ 14 \end{bmatrix}$$

$$y_1 = 8$$

$$y_2 = 11 - (3)(8) = -13$$

$$y_3 = 14 - (1)(8) - (-\tfrac{2}{5})(-13) = \tfrac{4}{5}$$

$$Y = \begin{bmatrix} 8 & -13 & \frac{4}{5} \end{bmatrix}^T$$

INSPIRING CREATIVE AND INNOVATIVE MINDS

UTM UNIVERSITI TEKNOLOGI MALAYSIA

www.utm.my

example

$$\begin{bmatrix} 1 & 2 & 1 \\ 0 & -5 & -2 \\ 0 & 0 & \frac{1}{5} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 8 \\ -13 \\ \frac{4}{5} \end{bmatrix}$$

$$x_3 = \frac{\frac{4}{5}}{\frac{1}{5}} = 4$$

$$x_2 = \frac{-13 - (-2)(4)}{-5} = 1$$

$$x_1 = \frac{8 - (2)(1) - (1)(4)}{1} = 2$$

$$X = \begin{bmatrix} 2 & 1 & 4 \end{bmatrix}^T$$

INSPIRING CREATIVE AND INNOVATIVE MINDS

UTM UNIVERSITI TEKNOLOGI MALAYSIA

www.utm.my

exercise

- Find the *LU* factorization of the matrix *A* using Doolittle form and use it to solve the linear system *AX*=*B*.

$$A=\begin{bmatrix}1 & 1 & 6\\ -1 & 2 & 9\\ 1 & -2 & 3\end{bmatrix}$$

(a) B=[7 2 10]^T (b) B=[23 35 7]^T

UTM UNIVERSITI TEKNOLOGI MALAYSIA

www.utm.my

Crout Factorization

- The diagonal elements of matrix *U* are 1's.

$$\begin{bmatrix}l_{11} & 0 & 0\\ l_{21} & l_{22} & 0\\ l_{31} & l_{32} & l_{33}\end{bmatrix}\begin{bmatrix}1 & u_{12} & u_{13}\\ 0 & 1 & u_{23}\\ 0 & 0 & 1\end{bmatrix}=\begin{bmatrix}a_{11} & a_{12} & a_{13}\\ a_{21} & a_{22} & a_{23}\\ a_{31} & a_{32} & a_{33}\end{bmatrix}$$

INSPIRING CREATIVE AND INNOVATIVE MINDS

UTM UNIVERSITI TEKNOLOGI MALAYSIA

www.utm.my

Example

- Find the *LU* factorization of the matrix using Crout form and use it to solve the linear system.

$$\begin{bmatrix} 7 & 2 & -3 \\ 2 & 5 & -3 \\ 1 & -1 & -6 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} -12 \\ -20 \\ -26 \end{bmatrix}$$

UTM UNIVERSITI TEKNOLOGI MALAYSIA

www.utm.my

Example

$$L = \begin{bmatrix} l_{11} & 0 & 0 \\ l_{21} & l_{22} & 0 \\ l_{31} & l_{32} & l_{33} \end{bmatrix} \qquad U = \begin{bmatrix} 1 & u_{12} & u_{13} \\ 0 & 1 & u_{23} \\ 0 & 0 & 1 \end{bmatrix}$$

INSPIRING CREATIVE AND INNOVATIVE MINDS

UTM UNIVERSITI TEKNOLOGI MALAYSIA

www.utm.my

Example

LU=A

$$\begin{bmatrix} l_{11} & 0 & 0 \\ l_{21} & l_{22} & 0 \\ l_{31} & l_{32} & l_{33} \end{bmatrix} \begin{bmatrix} 1 & u_{12} & u_{13} \\ 0 & 1 & u_{23} \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 7 & 2 & -3 \\ 2 & 5 & -3 \\ 1 & -1 & -6 \end{bmatrix}$$

UTM UNIVERSITI TEKNOLOGI MALAYSIA

www.utm.my

Example

$$\begin{bmatrix} l_{11} & 0 & 0 \\ l_{21} & l_{22} & 0 \\ l_{31} & l_{32} & l_{33} \end{bmatrix} \begin{bmatrix} 1 & u_{12} & u_{13} \\ 0 & 1 & u_{23} \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 7 & 2 & -3 \\ 2 & 5 & -3 \\ 1 & -1 & -6 \end{bmatrix}$$

$l_{11} = 7$ $\quad l_{21} = 2$ $\quad l_{31} = 1$

INSPIRING CREATIVE AND INNOVATIVE MINDS

UTM UNIVERSITI TEKNOLOGI MALAYSIA

www.utm.my

Example

$$\begin{bmatrix} 7 & 0 & 0 \\ 2 & l_{22} & 0 \\ 1 & l_{32} & l_{33} \end{bmatrix}\begin{bmatrix} 1 & u_{12} & u_{13} \\ 0 & 1 & u_{23} \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 7 & 2 & -3 \\ 2 & 5 & -3 \\ 1 & -1 & -6 \end{bmatrix}$$

$$u_{12} = \frac{2}{7} \qquad u_{13} = \frac{-3}{7}$$

UTM UNIVERSITI TEKNOLOGI MALAYSIA

www.utm.my

Example

$$\begin{bmatrix} 7 & 0 & 0 \\ 2 & l_{22} & 0 \\ 1 & l_{32} & l_{33} \end{bmatrix}\begin{bmatrix} 1 & 2/7 & -3/7 \\ 0 & 1 & u_{23} \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 7 & 2 & -3 \\ 2 & 5 & -3 \\ 1 & -1 & -6 \end{bmatrix}$$

$$l_{22} = 5 - (2)\frac{2}{7} = \frac{31}{7} \qquad \frac{31}{7}u_{23} = -3 - \frac{3}{7}(2)$$

$$u_{23} = -\frac{15}{31}$$

INSPIRING CREATIVE AND INNOVATIVE MINDS

UTM UNIVERSITI TEKNOLOGI MALAYSIA

www.utm.my

Example

$$\begin{bmatrix} 7 & 0 & 0 \\ 2 & 31/7 & 0 \\ 1 & l_{32} & l_{33} \end{bmatrix} \begin{bmatrix} 1 & 2/7 & -3/7 \\ 0 & 1 & -15/31 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 7 & 2 & -3 \\ 2 & 5 & -3 \\ 1 & -1 & -6 \end{bmatrix}$$

$$l_{32} = -1 - \frac{2}{7} = -\frac{9}{7} \qquad l_{33} = -6 + \frac{3}{7} - \frac{(-9)}{7}\frac{(-15)}{31} = \frac{-192}{31}$$

UTM UNIVERSITI TEKNOLOGI MALAYSIA

www.utm.my

Example

$$\begin{bmatrix} 7 & 0 & 0 \\ 2 & 31/7 & 0 \\ 1 & -9/7 & -192/31 \end{bmatrix} \begin{bmatrix} 1 & 2/7 & -3/7 \\ 0 & 1 & -15/31 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 7 & 2 & -3 \\ 2 & 5 & -3 \\ 1 & -1 & -6 \end{bmatrix}$$

INSPIRING CREATIVE AND INNOVATIVE MINDS

UTM UNIVERSITI TEKNOLOGI MALAYSIA

www.utm.my

Example

$$\begin{bmatrix} 7 & 0 & 0 \\ 2 & 31/7 & 0 \\ 1 & -9/7 & -192/31 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix} = \begin{bmatrix} -12 \\ -20 \\ -26 \end{bmatrix}$$

$$y_1 = -\frac{12}{7},$$

$$y_2 = -\frac{116}{31},$$

$$y_3 = \frac{902}{192}$$

UTM UNIVERSITI TEKNOLOGI MALAYSIA

www.utm.my

Example

$$\begin{bmatrix} 1 & 2/7 & -3/7 \\ 0 & 1 & -15/31 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} -12/7 \\ -116/31 \\ 902/192 \end{bmatrix}$$

$$x_3 = \frac{902}{192}$$

$$x_2 = \frac{-116}{31} + \frac{15}{31}\left(\frac{902}{192}\right) = \frac{-282}{192}$$

$$x_1 = \frac{-12}{7} - \frac{2}{7}\left(\frac{-282}{192}\right) + \frac{3}{7}\left(\frac{902}{192}\right) = \frac{138}{192}$$

UTM UNIVERSITI TEKNOLOGI MALAYSIA

www.utm.my

Exercise

- Find the *LU* factorization of the matrix *A* using Crout form and use it to solve the linear system *AX*=*B*.

$$A = \begin{bmatrix} 2 & 4 & -6 \\ 1 & 5 & 3 \\ 1 & 3 & 2 \end{bmatrix}$$

(a) B=[-4 10 5]^T (b) B=[20 49 32]^T